// TECHNICAL CASE STUDY

From Strava OAuth to Marathon Prediction.

A Growth & Tech white paper documenting OAuth integration, transparent estimation logic, and structured event tracking in a measurable product loop.

ARVID STENHAG — 2025

// 01 PROBLEM STATEMENT

The Problem.

Many runners collect extensive training data in Strava, yet lack a fast and frictionless way to translate that data into a credible marathon time estimate. Existing solutions are often hidden behind paywalls, require manual spreadsheet analysis, or involve complex setup.

From a growth perspective, the challenge is twofold: reduce activation friction in the OAuth flow and deliver immediate value to create a clear "wow moment" — while

OAuth flow and deliver immediate value to create a clear "wow moment" — while ensuring measurement is implemented from day one so iteration is data-driven.

// 02 HYPOTHESIS

The Hypothesis.

```
VALUE_EXCHANGE: "Connect Strava → Get marathon prediction instantly"
EXPECTED_OUTCOME: ↑ activation · ↑ repeat visits · ↑ referral behavior
MECHANISM: simplicity × transparency → trust → action
```

The simplicity of the promise is central: one click, authorize, return with a prediction.

// 03 ARCHITECTURE

System Architecture.

The solution consists of three integrated layers:

LAYER_01: LEAD MAGNET

Minimalistic frontend with a single primary CTA: 'Connect Strava.' Value proposition stated before permission request. Flow: click → authorize via OAuth → return with result.

LAYER_02: BACKEND API

Node + Express hosted on Render. Securely receives OAuth tokens, fetches activity summaries from Strava, calculates marathon estimate, returns structured JSON. Necessary to avoid exposing client secrets.

LAYER_03: CHROME EXTENSION

Manifest V3 activation helper and distribution layer. Demonstrates browser-based product enhancements and API communication within a browser environment.

// 04 ESTIMATION MODEL

Estimation Logic.

The estimation model is deliberately lightweight and transparent rather than algorithmically opaque. From Strava, the system focuses on recent runs — distance, duration, and inferred pace distribution — along with frequency and best-effort markers.

The model selects a representative benchmark run, normalizes pace to marathon distance, and applies a fatigue/decay factor for physiological drop-off at longer distances.

```
PREDICTION: 3:11:51
CONFIDENCE_BAND: ±6–10 min
MODEL: benchmark_pace × distance_factor × decay_coefficient
PRIORITY: explainability > complexity
```

// 05 INSTRUMENTATION

Event Tracking.

Instrumentation via Google Tag Manager connected to GA4. Events mapped directly to funnel stages:

EVENT	STAGE	SIGNAL
`lead_magnet_usage`	Awareness → Activation	CTA visibility & copy
`oauth_success`	Activation	Auth UX quality
`oauth_error`	Activation (drop-off)	Friction identification
`api_fetch_success`	Value Delivery	Backend reliability
`api_fetch_error`	Value Delivery (fail)	Infrastructure stability
`prediction_view`	Wow Moment	Core value confirmation
`whitepaper_download`	Engagement	Content depth interest
`plugin_repo_click`	Engagement	Technical credibility
`scroll_depth`	Content	Editorial engagement

Each event is linked to an optimization lever. High CTA clicks but low OAuth completion suggests authorization UX friction. High OAuth but frequent API errors indicates backend instability. Prediction views with low return rates suggest need for retention features.

Measurement is structurally integrated into iteration logic — not decorative.

// 06 GROWTH LOOP

The Growth Loop.

CONNECT → PREDICT → SHARE → RETURN ↺

Social sharing as referral trigger. Revisit after next long run as retention mechanic. Future comparison feature for lightweight virality through friend benchmarking.

// 07 UX PRINCIPLES

UX Design.

One primary CTA per section. Value proposition before permission request. Visible loading states. Post-prediction next steps: share result, re-run estimate, read model documentation. Activation leads into deeper engagement — never a terminal endpoint.

// 08 PRIVACY

Privacy & Permissions.

Least-privilege scopes only. No sensitive data requested. Client secrets never exposed on frontend. Clear messaging on data usage. Architecture is audit-friendly and compliant by design.

// 09 CAPABILITIES

What This Demonstrates.

✓ Measurable growth entry point
✓ OAuth + backend API integration
✓ Transparent estimation modeling
✓ Structured funnel-aligned tracking
✓ Browser extension as activation layer

Full-stack growth experiment — not a static prototype.

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// 10 ROADMAP

Future Directions.

→ Historical prediction tracking (progress visualization)
→ Race-goal inputs (sub-3, sub-3:30) with adaptive recs
→ Confidence calibration from training consistency
→ Premium analysis layer (opt-in, non-blocking)

Maintain frictionless core. Layer optional depth.

// CONCLUSION

Summary.

The Strava Marathon Connector is not merely a technical integration exercise. It is a structured demonstration of how Growth and Tech intersect: a clear value proposition, reduced activation friction, secure API architecture, explainable modeling, and measurable iteration. It showcases the ability to move from OAuth implementation to user-facing value in a controlled, data-driven product loop.

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